

When Accuracy Is Not Enough

Evaluating Semantic Fidelity in AI Systems

SFP 02 | Semantic Fidelity Project

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Abstract

AI evaluation often emphasizes factual accuracy, source support, and textual similarity. These measures are necessary, but they can rate an output highly after its organizing meaning has changed. This paper separates semantic fidelity from adjacent evaluation concepts and provides a practical method for locating representational loss. The method identifies the referent, the relationships that organize the source, the changes introduced by transformation, and the consequences of those changes for interpretation or action.

Core Claim

An output can preserve every important fact and still misrepresent the source by changing context, hierarchy, causality, uncertainty, or purpose.

The Evaluation Gap

Most evaluation methods inspect properties that are visible in the output. They detect unsupported claims, contradiction, omission, and broad semantic difference. These checks remain essential. They become insufficient when transformation changes the structure of meaning without producing an obvious error.

A summary can keep every stated fact and still make a tentative finding sound settled. Retrieval can surface relevant passages while dropping the qualification that controls how they should be read. An assistant can technically answer the request while drifting away from what the user was actually trying to accomplish. None of those failures necessarily produces a false statement, which is why conventional checks can miss them.

Evaluation Dimensions

Factual Accuracy

Primary question: Are individual claims correct?

Common blind spot: Correct claims can be arranged into a misleading account.

Source Faithfulness

Primary question: Does the source support the claims?

Common blind spot: Supported fragments can lose the source's structure.

Semantic Similarity

Primary question: Does the output resemble the source in meaning?

Common blind spot: Broad resemblance can hide changes in weight or implication.

Coherence

Primary question: Does the output hold together internally?

Common blind spot: A detached representation can remain coherent.

Semantic Fidelity

Primary question: Did the relationships organizing the meaning survive?

Common blind spot: Requires a defined referent and task.

What a Fidelity Test Examines

A semantic fidelity evaluation begins by identifying what the output is supposed to represent. Without a defined source, condition, intention, or outcome, fidelity cannot be distinguished from general quality.

The evaluator then identifies the relationships that organize the source. These may include causal order, uncertainty, exceptions, dependence, priority, scope, speaker position, or intended consequence. The test asks whether those relationships survived the transformation, not whether every sentence survived.

Finally, the evaluator considers the effect of the changes. If a reader would form a materially different interpretation, make a different decision, or lose the ability to reconstruct the original position, the loss is significant even when the output remains accurate at the claim level.

A Five-Step Diagnostic

A fidelity check begins with the referent. The evaluator first has to establish what the output is supposed to remain connected to, then identify the relationships, conditions, and uncertainties that organize the source. From there, the task is to look at what changed during transformation. What was retained, removed, flattened, resolved, or given different weight?

The next question is whether those changes matter. Would the transformed version lead to a different interpretation or action? Finally, the evaluator should ask whether the missing structure can be recovered and, if it is recovered, whether that correction changes what the system does next.

The point is not to collapse fidelity into a universal score. Different tasks depend on different relationships. The diagnostic is meant to make the basis of judgment visible and prevent surface correctness from standing in for representational integrity.

Characteristic Failure Patterns

Fidelity loss tends to show up in a handful of recurring ways. Important distinctions can be flattened until primary claims, supporting evidence, and peripheral examples appear equally important. Qualified claims can harden into direct statements, while sequence or correlation can quietly turn into causal explanation. Accurate material can also be moved into a new context where it carries a different meaning, or a system can satisfy the literal form of a request while drifting away from its underlying purpose.

What makes these failures difficult to notice is that they often look like improvements. The output may become shorter, cleaner, more decisive, or easier to evaluate. What disappears is the structure that made the original reliable.

From Local Scores to Transformation Chains

AI systems rarely perform a single transformation. Retrieval selects material, ranking establishes prominence, summarization compresses it, generation reconstructs prose, and memory carries part of the result forward. Each stage can appear acceptable when tested alone. Semantic loss can still accumulate across the chain.

Evaluation should therefore compare the final output with the original referent whenever possible. When direct comparison is impossible, provenance should remain recoverable so that later users can distinguish contact with a source from agreement among derivative representations.

Design Implications

Fidelity-aware systems should preserve source access, expose consequential omissions, retain uncertainty markers, distinguish generated reconstruction from retrieved material, and allow correction to update later behavior. Evaluation sets should include cases where every statement is technically defensible but the overall interpretation is wrong.

The central question is not whether the model produced acceptable language. It is whether the transformation preserved the relationships required for the output to remain a reliable representation.

Conclusion

Accuracy identifies false statements. Semantic fidelity identifies altered meaning that remains plausible because the statements still look correct. As AI systems mediate longer chains of representation, this distinction becomes necessary for detecting drift before visible failure appears.

